ADET Prelims

Application Development and Emerging Technologies (Our Lady of Fatima University)



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ARDUINO MICROCONTROLLER

What Is Arduino?

- It is an **open-source electronics platform/microcontroller.**
- Also, has programming its own software/IDE.
- Designed to be less expensive and easy to use and learn.
- Can be connected to many relays, motors, sensors, etc.

WHAT IS THE USE OF ARDUINO?

- For everything from robots, tiny weather display to a laser harp. The Arduino can be used as the brains behind almost any electronics project.

HISTORY

- Development started in Interaction Design Institute Ivrea in Ivrea, Italy.
- Based on a thesis of a Columbian Student named Hernando Barragan.
- The thesis is named "Wiring: Prototyping Physical Interaction Design".
- Wiring's objective was to "make it easy for artists and designers to work with electronics".
- First prototype was produced on 2005 by Massimo Banzi.
- He was later on joined by four others; David Cuartielles, Tom Igoe, Gianluca Martino, and David Mellis.

NOTE:

1. THESIS = Columbian Student named Hernando Barragan (named "Wiring: Prototyping Physical Interaction Design")
2. 1ST PROTOTYPE = Massimo Banzi.
3. Later on, Joined by four others; David Cuartielles, Tom Igoe, Gianluca Martino, and David Mellis.

Different Types of Arduinos

1. **Arduino Uno R3**

- Arduino UNO is a basic and inexpensive Arduino board and is the most popular of all the Arduino boards with a market share of over 50%.
- Arduino UNO is considered to be the best prototyping board for beginners in electronics and coding.

2. **Lilypad Arduino**

- The LilyPad Arduino is used in e-textiles and wearable projects.
- It can be sewn to fabric and similarly mounted power supplies, sensors and actuators.

3. **Arduino Nano**

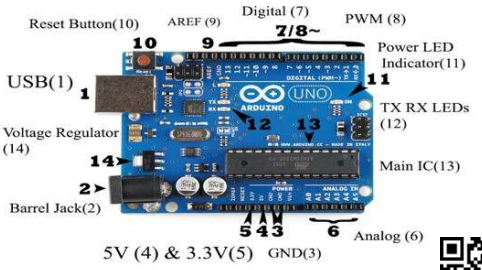
- The Nano is similar to the UNO in terms of functionality but it has slightly different components and layout.
- It's most defining characteristic is the lack of a DC power jack, which explains its compact size.

4. **Arduino Esplora**

- The Esplora allows users more direct control over their creations.
- This board is more used as an user interface for other electronic projects.

PARTS OF ARDUINO

1. Power (Usb / Barrel Jack)
2. Pins
3. Reset Button
4. Power Led Indicator
5. Tx Rx Leds
6. Main Ic
7. Voltage Regulator (14)



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Power (USB / Barrel Jack)

- It can be powered via a USB connection (1) or a DC power source via the jack (2).

PINS

- GND (3): 2 Ground Pins, used to ground your circuit.
- 5V(4) & 3.3V(5): Supplies power with 5 or 3.3 volts.
- Analog (6): These pins can read the signal from an analog sensor(i.e. Temperature sensor).
- Digital (7): These pins can be used for both digital input and output.
- PWM (8): These pins act as normal digital pins, but can also be used for something called Pulse-Width Modulation (PWM).
- AREF (9): Stands for Analog Reference. Most of the time you can leave this pin alone. It is sometimes used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins.

Reset Button

- Pushing it will temporarily connect the reset pin (10) to ground and restart any code that is loaded on the Arduino.

Power Led Indicator

- To the right of the word “UNO”, there’s a tiny LED next to the word ‘ON’ (11). This LED should light up whenever you plug your Arduino into a power source.

TX RX Leds

- TX is short for transmit, RX is short for receive. It is located to the left of the word “Arduino” (12).
- These LEDs will give us some nice visual indications whenever our Arduino is receiving or transmitting data.

Main IC

- The black thing with all the metal legs is an IC, or Integrated Circuit (13). This is the “brain” of the Arduino.
- You may need to know the IC type (along with your board type) before loading up a new program from the Arduino software. This information can usually be found in writing on the top side of the IC.

Voltage Regulator (14)

- It controls the amount of voltage that is let into the Arduino board. It will turn away an extra voltage that might harm the circuit.
- It has its limits, usually anything greater than 20 volts.

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Electronics Principles for Microcontrollers

In developing an Arduino project or any projects that uses a microcontroller, electronic components are quite important. This means that in order to build an electronic project, we need to identify and understand first some of the basic electronic components.

RESISTORS

- A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element.

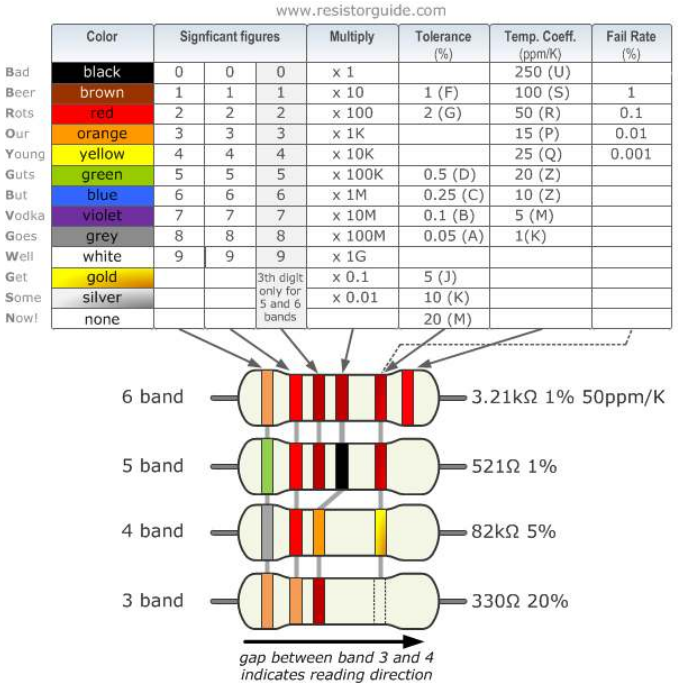
BRIEF HISTORY ON RESISTORS

- Georg Ohm was a German physicist and mathematician
- He was known for his “Ohm’s Law”
- Ohm’s Law states that the current through a conductor between two points is directly proportional to the voltage across the two points.
- The resistor was invented by Otis Frank Boykin on 1959
- He also invented a control unit for the artificial cardiac pacemaker
- He died because of heart failure on 1982

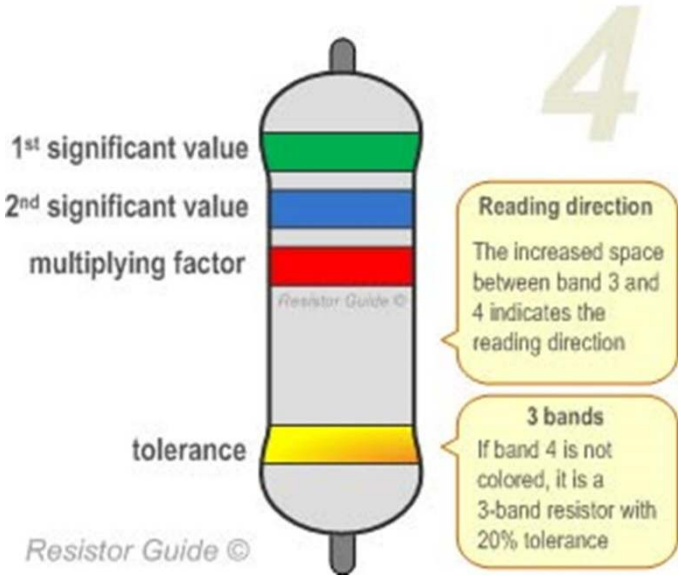
TYPES OF RESISTORS



STANDARD RESISTOR COLOR CODE



HOW TO READ A RESISTOR?



CAPACITORS

- A Capacitor is an electronic component that stores electric charge.

WHAT IS A CAPACITOR?

- The capacitor is made of two close conductors (usually plates) that are separated by a dielectric material. The plates accumulate electric charge when connected to a power source. One plate accumulates positive charge and the other plate accumulates negative charge.

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- The capacitors are measured in Farads.

TYPES OF CAPACITORS



TRANSISTORS

- A Transistor is a three-leg insect type component, which is placed singly in some devices but in computers it is packed inside in millions of numbers in small microchips.

BRIEF HISTORY ABOUT TRANSISTORS

- Before transistors, first generation computers used vacuum tubes for logic circuitry.
- The world's first general-purpose electronic computer, ENIAC, had more than seventeen thousand vacuum tubes
- Vacuum tube cathodes are needed to be heated first to work. Because of these vacuum tubes are burnt out regularly and needed to be replaced.
- Because of the large number of vacuum tubes, the ENIAC was very power hungry
- The first transistors were developed by a group of scientists led by John Bardeen, Walter Brattain, and William Shockley at AT&T Bell Laboratories in mid 1940s

INTEGRATED CIRCUITS

- An integrated circuit, or IC, is a small chip that can function as an amplifier, oscillator, timer, microprocessor, or even a computer memory

TYPES OF IC'S

- Digital Integrated Circuits
- Analog Integrated Circuits
- Mixed Integrated Circuits
- Logic Circuits
- Comparators
- Switching ICs
- Audio amplifiers
- Operational amplifiers

POTENTIOMETERS

- A potentiometer is a three-terminal resistor with a sliding or rotating contact that forms an adjustable voltage divider.

WHAT IS A POTENTIOMETER?

- A potentiometer is a manually adjustable, variable resistor with three terminals. Two terminals are connected to a resistive element, the third terminal is connected to an adjustable wiper. The position of the wiper determines the output voltage.

DIFFERENT TYPES OF POTENTIOMETERS

- There are two major types of potentiometers, Rotary potentiometers and Linear Potentiometers
- **Rotary potentiometers** – The most common type of potentiometer where the wiper moves along a circular path.



Dual-gang potentiometer *Concentric potentiometer* *Multi-turn potentiometer*

- **Linear potentiometers** - Potentiometers where the wiper moves along a linear path. Also known as slider, slide pot or fa

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Slide potentiometer



Motorized fader



Multi-turn linear trimpot

DIODES

- A diode is a two-terminal electrical device that allows the transfer of current in only one direction.

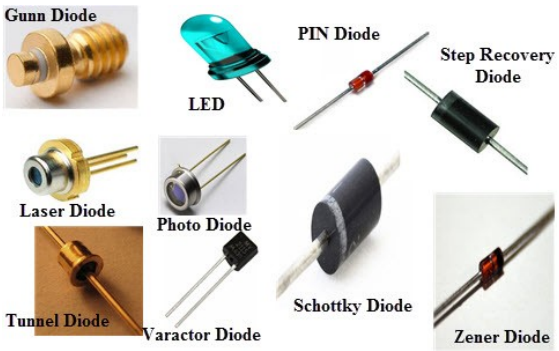
WHAT IS A DIODE?

- A diode is used for rectifying waveforms, within radio detectors or within power supplies.
- They can also be used in various electrical and electronic circuits where 'one-way' result of the diode is required.
- Most of the diodes are made from semiconductors like Si (silicon), but sometimes, Ge (germanium) is also used.

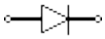
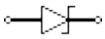
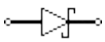
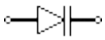
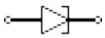
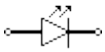
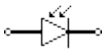
BRIEF HISTORY ABOUT DIODES

- The story of invention revolves around the perseverance of an American Physicist Russel S. Ohl who discovered the basic form of PN Junction while his research at Bell Laboratories.
- Russel Ohl (aka R.S Ohl) was a researcher with AT &T Bell laboratories who specialized in materials research.

TYPES OF DIODES



DIFFERENT DIODE SYMBOLS

Symbol	Name	Description
	Diode	Diode allows current flow in one direction only (left to right).
	Zener Diode	Allows current flow in one direction, but also can flow in the reverse direction when above breakdown voltage
	Schottky Diode	Schottky diode is a diode with low voltage drop
	Varactor / Varicap Diode	Variable capacitance diode
	Tunnel Diode	
	Light Emitting Diode (LED)	LED emits light when current flows through
	Photodiode	Photodiode allows current flow when exposed to light

LIGHT-EMITTING DIODES (LED)

- An LED is a small component that gives light.

HOW AN LED WORKS?

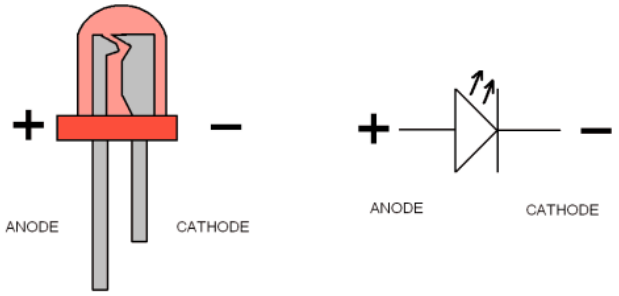
- To understand the workings of a light-emitting diode, you need to get into the physics part of semiconductors. Simply explained, the LED is similar to a standard diode, but with the addition of a phenomenon called Electroluminescence.

CONNECTING AN LED

- The LED has two pins called anode and cathode.

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- The anode is the longest pin. This is the pin you connect to the most positive voltage. The cathode is the pin you connect to the most negative voltage. They must be connected correctly for the LED to work.



LIGHT DEPENDENT RESISTORS (LDR)

- An LDR is a component that has a (variable) resistance that changes with the light intensity that falls upon it. This allows them to be used in light sensing circuits.

WHAT IS AN LDR?

- Light dependent resistors or photoresistors, are sensitive devices most often used to indicate the presence or absence of light, or to measure the light intensity.

TYPES OF LDR



SOME APPLICATIONS OF LDRS

- There are many applications for Light Dependent Resistors. These include:
  - **Lighting switch**
    - The most obvious application for an LDR is to automatically turn on a light at a certain light level. An example of this could be a street light or a garden light.
  - **Camera shutter control**
    - LDRs can be used to control the shutter speed on a camera. The LDR would be used to measure the light intensity which then adjusts the camera shutter speed to the appropriate level.

BUTTON SWITCH

- A Button is a simple switch mechanism for controlling some aspect of a machine or a process.

WHAT IS A BUTTON SWITCH?

- Buttons are typically made out of hard material, usually plastic or metal. The surface is usually flat or shaped to accommodate the human finger or hand, so as to be easily depressed or pushed.
- Buttons are most often biased switch, although many un-biased buttons (due to their physical nature) still require a spring to return to their un-pushed state.

TYPES OF BUTTON SWITCH



RELAYS

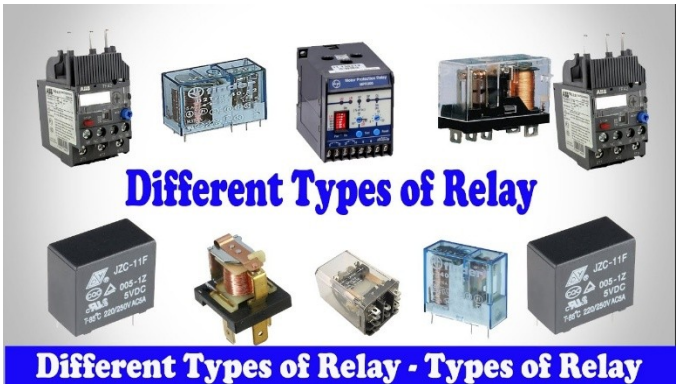
- Relays are switches that open and close circuits electromagnetically or electronically

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SOME TYPES OF RELAYS

- A type of relay that can handle the high power required to directly control an electric motor or other loads is called a contactor.
- Solid-state relays control power circuits with no moving parts, instead using a semiconductor device to perform switching.
- Magnetic latching relays require one pulse of coil power to move their contacts in one direction, and another, redirected pulse to move them back. Repeated pulses from the same input have no effect. Magnetic latching relays are useful in applications where interrupted power should not be able to transition the contacts.

TYPES OF RELAYS



MULTIMETERS

- A multimeter is also known as a volt-ohm meter, is a handheld tester used to measure electrical voltage, current(amperage), resistance, and other values.

TYPES OF MULTIMETERS

- Analog multimeters use a microammeter with a moving pointer to display readings.
- Digital multimeters (DMM, DVOM) have a numeric display, and may also show a graphical bar representing the measured value.

Key Terminologies and Important Names in Arduino Microcontroller:

1. Arduino Microcontroller:

- **Definition:** An open-source electronics platform used to build digital devices and interactive objects, based on a microcontroller (such as ATmega328).
- **Uses:** Can control a variety of electronic components like motors, sensors, and relays. It's used in various projects like robots, weather displays, and interactive art.

2. Key Names and Individuals:

- **Hernando Barragan:** Colombian student who authored the thesis "Wiring: Prototyping Physical Interaction Design," which inspired the development of Arduino.
- **Massimo Banzi:** One of the founders of Arduino, who produced the first prototype in 2005.
- **David Cuartielles, Tom Igoe, Gianluca Martino, and David Mellis:** The other four individuals who contributed to the Arduino project after Massimo Banzi.

3. Types of Arduino Boards:

- **Arduino Uno R3:** Most popular, basic board for beginners.
- **Lilypad Arduino:** Used for e-textiles and wearable tech; can be sewn into fabrics.
- **Arduino Nano:** A compact version of the UNO with no DC power jack.
- **Arduino Esplora:** A user interface board for creating interactive projects.

4. Arduino Components:

- **Power (USB / Barrel Jack):** For powering the Arduino board.
- **Pins:** Including GND (ground), 5V & 3.3V (power), analog (input), digital (input/output), PWM (pulse-width modulation), and AREF (analog reference).
- **Reset Button:** Restarts the program loaded in the Arduino.
- **TX RX LEDs:** Indicate data transmission and reception.
- **Main IC:** The Integrated Circuit that functions as the brain of the Arduino.
- **Voltage Regulator:** Protects the circuit from excess voltage.

5. Electronics Principles for Microcontrollers:

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- **Resistors:** Limit current flow in circuits (Ohm's law by Georg Ohm).
- **Capacitors:** Store electrical charge, measured in Farads.
- **Transistors:** Act as amplifiers or switches in circuits.
- **Integrated Circuits (ICs):** Compact chips that function as amplifiers, microprocessors, and more.
- **Potentiometers:** Adjustable resistors with a sliding or rotating contact.
- **Diodes:** Allow current flow in only one direction; used for rectification.
- **Light-Emitting Diodes (LEDs):** Emit light when current flows through them.

6. Types of Electronic Components:

- **Resistor:** Passive component that resists the flow of current.
- **Capacitor:** Stores electric charge for later use in circuits.
- **Transistor:** Acts as a switch or amplifier.
- **Integrated Circuits (ICs):** Small chips that perform a variety of functions.
- **Diode:** A two-terminal device that restricts current to flow in only one direction.
- **LED:** A type of diode that emits light when current flows through it.

7. Important Diodes and Their Types:

- **Zener Diode:** Allows reverse current once it exceeds a specific breakdown voltage.
- **Schottky Diode:** Known for low voltage drop.
- **Photodiode:** Generates current when exposed to light.
- **LED (Light-Emitting Diode):** Emits light when current flows through.

8. Sensors and Components:

- **LDR (Light Dependent Resistor):** A resistor whose resistance decreases with increasing light intensity.
- **Button Switch:** A mechanical switch to open or close circuits.
- **Relay:** An electromagnetic switch used to control high-power devices with low-power circuits.

9. Types of Relays:

- **Contactor:** A relay that controls electric motors or large power loads.
- **Solid-State Relay:** Uses semiconductor components without moving parts.
- **Magnetic Latching Relay:** Uses pulse signals to change its state without power.

10. Measurement Tools:

- **Multimeter:** A device used to measure electrical values such as voltage, current, and resistance.
    - **Analog Multimeter:** Uses a moving needle to display readings.
    - **Digital Multimeter:** Uses a digital display for precise measurements.

Key Details:

- **Arduino Software/IDE:** The programming interface for writing and uploading code to the Arduino board.
- **Common Applications:** Robotics, sensor networks, home automation, wearable tech, and interactive art.
- **Arduino's Accessibility:** Designed to be beginner-friendly, making it a popular choice for DIY electronics projects.

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ADET ACTIVITY SKETCH

ULTRASONIC SENSOR

```
#include <Ultrasonic.h>
Ultrasonic ultrasonic(12, 13);
int distance;

void setup() {
  Serial.begin(9600);
}

void loop() {
  distance = ultrasonic.read();
  Serial.print("Distance in CM: ");
  Serial.println(distance);
  delay(1000);
}
```

SERVOMOTOR

```
#include <Servo.h>
int servoPin = 7;
int servoPos = 0;
Servo myServo;

void setup() {
  Serial.begin(9600);
  myServo.attach(servoPin);
}

void loop() {
  Serial.println("Input a value");
  while (Serial.available() == 0) {
  }
  servoPos = Serial.parseInt();
  myServo.write(servoPos);
}
```

LED

```
int led1 = 4;
int led2 = 5;

void setup() {
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop() {
  digitalWrite(led1, HIGH);
  delay(1000);
  digitalWrite(led1, LOW);
  digitalWrite(led2, HIGH);
  delay(1000);
  digitalWrite(led2, LOW);
}
```

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